# GAZE — CP2 Progress Report

CSST 106 · 22 October 2026 · Team submission

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## Project and current outcome

GAZE supports instructor review of visible head movements during classroom examinations. The working CPU prototype processes images and saved video, annotates faces and directions, and can save seat histories and rule-based movement records. Head movement alone does not establish misconduct. Live input, integrated event review, and representative classroom evaluation remain unfinished.

## Data, baseline, and measured result

The selected Pointing04 subset contains 360 images from 15 people (Gourier et al., 2004): 240 validation images from ten people and 120 reserved test images from five different people. All images are 384 × 288 pixels. Twenty demonstration examples belong to the validation split. Training is not applicable because no project-specific classifier is trained.

MediaPipe supplies pretrained landmarks (Google, 2026). Shared image-preparation code feeds a geometric direction baseline and an approximate OpenCV pose comparison (OpenCV, n.d.). Results below use the same 240 validation images; missed or unknown predictions count against accuracy.

| Method | Correct / 240 | Accuracy | Macro-F1 |
|---|---|---|---|
| Initial geometric baseline | 166 | 69.17% | 0.6339 |
| Validation-selected baseline | 208 | 86.67% | 0.8798 |
| Approximate pose comparison | 162 | 67.50% | 0.6734 |

The selected baseline produced six unknown results. Its settings were adjusted using validation data, so this is a development measurement, not a final held-out score. The reserved test set has not been scored in the reviewed results. Source: GAZE development measurements, October 2026.

## Completed versus pending work against the proposal timeline

| Checkpoint / target | Completed work | Pending work |
|---|---|---|
| CP1 · 15 Oct | Permitted samples, labels, landmarks, annotations, contact sheet; fresh-clone five-image spike. | Final proposal, member contributions, pitch and earlier hand-in evidence. |
| CP2 · 22 Oct | Pinned environment, shared preprocessing, image command, video path, validation measurements and saved outputs. | Another member’s intended-laptop run and classroom code explanations. |
| CP3 · 29 Oct | Approximate pose method, histories and method comparison. | Labeled video intervals and measured seat consistency. |
| CP4 · 12 Nov | Duration/repetition rules and adjustable settings. | Live input, integrated event review, event errors and laptop performance. |
| CP5 / defense · 19 Nov | Planning and a reserved test set. | Frozen final evaluation, final rehearsal, report, presentation and individual defenses. |

The fresh-clone spike installed a new pinned environment, downloaded and checked the model, and saved five new annotated outputs. One direction was wrong and one was unknown; those failures are retained. A single public-domain FourPeople office clip supports the video demonstration (Xiph.Org Foundation, n.d.); it does not establish classroom event accuracy.

# Progress, blockers, and revised plan



## Effort-weighted completion: 50.0%

Completed effort / planned effort × 100 = 69 / 138 × 100 = 50.0%. The calculation covers the complete roadmap, including later evaluation, documentation, demonstrations, and three individual defenses. The accompanying worksheet maps all 40 tasks. Two administrative tasks have zero delivery weight; 18 acceptance-verified work tasks receive credit. Unfinished tasks receive zero, including partly implemented work and the unfinished hand-in/class-review task.

Weights were estimated retrospectively for reporting: 1 point for a small review step; 2 for a small utility or artifact; 3 for a bounded component, verification, or document; 5 for substantial algorithm, data, or evaluation work; and 8 for complex integration. They are estimates, not logged hours or course marks. Reused work is not counted twice. The weights are unchanged from the previous calculation; verified task #8 added two points.

| Roadmap area | Planned | Completed | Remaining |
|---|---|---|---|
| CP1 | 24 | 17 | 7 |
| CP2 | 36 | 31 | 5 |
| CP3 | 24 | 13 | 11 |
| CP4 | 31 | 8 | 23 |
| CP5 | 14 | 0 | 14 |
| Individual defenses | 9 | 0 | 9 |
| Total | 138 | 69 | 69 |

## Blockers and revised plan

| Target | Next action / blocker |
|---|---|
| 23 Oct class check | Another member reproduces the README on the intended laptop; confirm personal contributions and explain code each member wrote. Earlier proposal and hand-in evidence also needs review. |
| Toward 29 Oct (CP3) | Obtain suitable permitted footage; check hand counts on two to three clips; label seat/movement/writing/gap intervals and measure seat consistency. |
| Toward 12 Nov (CP4) | Complete live input and integrated event review; score real movement events and writing false alerts; benchmark the intended laptop. |
| Toward 19 Nov (final) | Freeze settings before reserved-test evaluation; rehearse the supported demonstration and backup; complete final reporting and individual defenses. |

These are proposed work targets within the original course schedule, not deadline extensions. The CP2 hand-in deadline remains Thursday 22 October, 11:30 PM. The principal gaps are representative classroom labels, camera/face calibration assumptions, and human/laptop verification. The working saved-video direction display is the documented fallback.

Recorded implementation is attributed to Jeremy Garin with coding-agent assistance. Marc Justin Alberto's and Jose III Aquino's personal implementation contributions and explanations remain to be confirmed. The 50.0% estimate describes project delivery under the stated weights; instructor acceptance and classroom accuracy remain separate.

## References

Google. (2026, October 9). *Face landmark detection guide*. Google AI Edge. https://developers.google.com/edge/mediapipe/solutions/vision/face_landmarker

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OpenCV. (n.d.). *Perspective-n-Point (PnP) pose computation*. https://docs.opencv.org/4.11.0/d5/d1f/calib3d_solvePnP.html

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